

CYBER SECURITY FUNDAMENTALS NOTES

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Cyber Security refers to the technologies, processes and practices designed to protect networks, devices, app and data from any kind of cyber-attacks. Cyber security may also known as information technology (IT) security.

Cyber Security is all about protecting your devices and network from unauthorized access or modification. The Internet is not only the chief source of information, but it is also a medium through which people do business.



Cybersecurity Definition:

Today, people use the Internet to advertise and sell products in various forms, communicate with their customers and retailers, and perform financial transactions. Due to this, hackers and cybercriminals use the internet as a tool to spread malware and carry out cyber-attacks.

Cybersecurity aims to protect the computers, networks, and software programs from such cyber-attacks. Most of these digital attacks are aimed at accessing, altering, or deleting sensitive information; extorting money from victims; or interrupting normal business operations.

Types of Cybersecurity

Cyber Security is classified into the following types:

#Information Security

Information security aims to protect the users' private information from unauthorized access, identity theft. It protects the privacy of data and hardware that handle, store and transmit that data. Examples of Information security include User Authentication and Cryptography.

#Network Security

Network security aims to protect the usability, integrity, and safety of a network, associated components, and data shared over the network. When a network is secured, potential threats gets blocked from entering or spreading on that

network. Examples of [Network Security](#) includes Antivirus and Antispyware programs, Firewall that block unauthorized access to a network and VPNs (Virtual Private Networks) used for secure [remote access](#).

#Application Security

Application security aims to protect software applications from vulnerabilities that occur due to the flaws in application design, development, installation, upgrade or maintenance phases.

Types of Cybersecurity Threats

There are many different types of cybersecurity threats, some of the most common types of threats are listed below,

Viruses

Viruses are a type of malware programs that are specially designed to cause damage to the victims' computer. Viruses can self-replicate under the right conditions and can infect a computer system without the permission or knowledge of the user.

It has two major characteristics, the ability to replicate itself and the ability to attach itself to another computer file. A virus has the capability to corrupt files and steal private information like credit card details of the user and send them back to the hacker.

Viruses cannot exist on its own, i.e., without a host program; it is usually present as a parasite on another program. Piggybacking on another program allows the virus to trick users into downloading and executing it.

When a virus-infected program is executed, the virus also gets executed. Once executed, malware virus performs two primary functions simultaneously: Replicate and Infect.

The virus takes control of the host computer and begins searching for other programs on the same or other disks that are currently uninfected. When it finds one, it then copies itself into the uninfected program.

After replicating itself into many copies and infecting other uninfected programs, host program returns to its original form. When the host program gets terminated by the user, the virus too will stop replicating. Since all these activities occur in the background, the user will be completely unaware of the virus.

Some viruses will remain active in the system memory even after the user terminated the host program. This type of virus will stay in system memory until the computer is turned OFF. The next time the user boots his computer system, he/she might unknowingly execute one of the infected applications on the computer.

When the virus remains active in the system memory, it may deliver the payload. The payload can be anything from deleting files or slowing down the computer. It could modify data files, damage or delete data files and programs.

Identity Theft

It is a type of cybersecurity threat which involves the stealing of personal information of the victims from social media websites such as Facebook, Instagram, etc. and using that info to build a picture of the victims. If sufficient sensitive information is gathered it could allow the cybercriminal to pretend as you in some way.

In some cases, hackers may steal the bank details of the victims and use it for their personal gain.

Password Attacks

It is a type of *cybersecurity* threat which involves a hacking attempt by hackers to crack the passwords of the user. With the help of a hacking tool, hackers may enter many passwords a second to crack the victim's account credentials and gain access. Hackers may also perform password attacks on a computer login screen to gain access to a victim's computer and the data stored in it.

Spyware and Keyloggers

Malware such as the spyware can spy on computing habits of the victims. Some malware such as the keyloggers can record the victims' keystrokes including their passwords, PIN numbers, and credit card details. Keyloggers and spyware programs enter the victims' system when they download and install seemingly benign software from a dubious website.

Spyware and keyloggers gather user information, passwords, browsing history, etc., and then transmits them to its creators (hackers) who may sell or distribute this

personal information to third parties. Hackers may also use that information to steal money from the victim's bank accounts.

Adware

Adware is a group of malware that is known to generate these pop-ups. If a user notices strange pop-up messages on their computer screen, it is most likely to be a malware attack. The main intention of adware is to gain permissions that will then allow them to install additional malicious software. If the user downloads that additional software, it may then either delete or steal your data. Some of these pop-up messages can also be used to simply bombard your computer screen with unwanted information such as advertisements.

Trojans

Trojans are a type of malware programs that disguise themselves as harmless or useful software. Trojans can cause a variety of malicious activities on the victims' computer including downloading malicious programs, deleting or stealing files and providing hackers unauthorized access to the victims' computer.

Ransomware

Ransomware is a group of malware which locks or encrypts the victim's computer and demands payment for decrypting the computer. The primary motive for all ransomware attacks is always monetary.

Unlike many other types of cyber attacks, ransomware attacks notify the victim about the exploit, and it also gives instructions on how to recover from it (usually it demands payment for recovery). To avoid a crackdown by law enforcement, hackers who are behind the ransomware attacks typically demand payments in virtual currencies, such as the Bitcoins.

Since ransomware is one of the most prominent and widespread among all other cyber threats, let's have a closer look at how it operates.

Ransomware: Infection Mechanism

Ransomware infects a computer through various means such as through malicious email attachments, malicious links in shady websites. Most ransomware attacks are based on remote desktop protocol and other tactics that don't rely on user interaction.

Users may inadvertently download ransomware when they visit compromised websites. Ransomware malware can also piggyback on other malicious software applications as a payload. Some ransomware variants are known to spread through email attachments from malicious emails or released by exploit kits onto vulnerable computers.

Once the ransomware gets executed, it can change the victim's login credentials, encrypt files and folders on the victim's device, as well as on other connected devices.

In the first case scenario (changing the login credentials), ransomware shows a full-screen image or notification on the

infected system's screen, which cannot be closed at the user's will. It may also have the instructions on how users can pay for the ransom and get the decryption key.

In the second case scenario (encrypting files and folders), the ransomware malware prevents access to valuable files like documents and spreadsheets.

Browser Hijacker

Some malicious software such as the Browser Hijacker redirects the victims' browser to specific websites that are chosen by the hacker or to a site that pays the hacker based on the number of hits it receives. In some cases of scareware infections, the entire root drive of the victims and all of their subdirectories will be hidden. It may also record their personal information and transmit it to the hacker.

Zero-Day Attacks

Zero-day attacks are carried out using zero-day malware. This zero-day malware exploits a previously unknown vulnerability that has not been addressed or patched. Since the zero-day vulnerability is previously not known, the zero-day exploits often occur without the consent of the users as there will be no patches available at the time of infection.

Phishing Emails

Phishing emails are intended to steal private user information like user login credentials and credit card numbers. It is a type of social engineering attack used by hackers wherein the user is tricked into clicking malicious

attachments or links that download malware. Since phishing attacks use seemingly benign emails or software, it becomes difficult for the users to ascertain them.

Phishing emails are generally used for stealing private information from the users whereas spam emails are generally used to flood the Internet with numerous copies of the same message, in an attempt to force the message on computer users who would not otherwise choose to receive it.

Where do Cybersecurity Threats Originate?

There are many different ways by which cyber threats infects a victim's computer, some of the most common ways are listed below.

#Shady Websites

Hackers use seemingly legitimate software and websites to lure users into downloading malware. The lure of free of free money or games entices some users. Even people who do not visit these shady sites are prone to enticing links being hidden on their computers.

Malware that originates from these sites may travel around the internet and land on an innocent users computer redirecting their browsing experience to these sites.

#Peer to Peer File Sharing

Peer to Peer (P2P) file-sharing networks is one of the most popular ways to share movies, games, music, and other files online. In a typical P2P network, participants make a portion

of their own computing resources available to other network participants.

In essence, file sharing over a P2P network allows computer users to share files directly from the computers of each other. P2P file sharing is also a very commonly used method for distributing malware and performing other malicious deeds.

#Torrent Downloads and Phishing Emails

Trying to find a particular movie that is still in theaters? Maybe you want a free copy of the latest PC Game. Torrent sites are used by computer savvy users that have malware removal in their daily agenda.

Keygens, Cracks, Serial Coders, all of these might be what you need but don't be surprised when you are infected. In most cases, the file you are downloading could be a rogue malware installer written by a savvy programmer.

Email is the breeding ground for many malware. If you open a phishing email that is sent with an attachment, it instantly collects information within your email, mainly your address book. It will immediately send similar phishing emails to all of the contacts in your address book, spreading the problem.

Why is Cybersecurity Important?

The infamous cyber attacks such as the GoldenEye and WannaCry ransomware attacks have crippled several organizations and forced many to shut down their operations. In the wake of these sophisticated cyber attacks and security breaches, cybersecurity has taken the spotlight among organizations of all sizes.

New variants. New tactics. Cyber threats continue to evolve. Not only have we seen an increase in cyber attacks on businesses and individuals, but the level of sophistication in those cyber attacks have also increased, as well.

In the years to come, there will be even more advanced cyber attacks using new technologies, victims, and intentions. There will be a dramatic rise in the availability of Ransomware-as-a-Service and Malware-as-a-Service on the dark web. It will allow anyone, no matter their technical knowledge, to easily and quickly initiate a cyber attack.

Nevertheless, owing to the extent of damages caused by cyber attacks in the past, now there is a much greater awareness about the cyber attacks and the need for better cybersecurity measure among organizations of all types. This will serve as a motivation for cybercriminals to up their game by staging new and more sophisticated attacks in the future.

Wide-ranging security vulnerabilities, faster and more sophisticated cyber attacks are all making it extremely difficult for security experts to prevent those threats. Thus,

there should be a proper cybersecurity plan in place to prevent cyber attacks from causing any damage.